

Aim:

Write a Python program that demonstrates the use of user-defined functions. Your program should perform the following tasks:

- Define a function that takes two integers a and b as parameters and returns their sum.
- Define a function that calculates and returns the area of a circle using the formula: $\text{Area} = \pi \times r^2$, where the r is the radius (a floating-point value).
- Define a function that takes a string s and returns its reverse.
- Define a function that takes an integer n and returns both its square and square root (rounded to 2 decimal places).
- Define a function that demonstrates default arguments and keyword arguments, by constructing and returning a greeting message in the format:

```
<greeting>, <name>!
```

Input Format:

The input should be entered in the following order, each on a new line:

```
<a> <b> //integer values of two numbers
<radius>
<string>
<integer>
<name>
<greeting>
```

Output Format:

Print the results of all operations in a single line per result:

```
Sum of <a> and <b> is: <result>
Area of circle with radius <r> is: <area>
Reverse of 's' is: <reversed_string>
Square of <n> is <square>, Square root is <sqrt>
<greeting_message_1>
<greeting_message_2>
<greeting_message_3>
```

- **greeting_message_1** uses default arguments

```
Hello, User!
```

- **greeting_message_2** uses the keyword argument name only

```
Hello, <name>!
```

- **greeting_message_3** uses keyword arguments for both name and greeting

```
<greeting>, <name>!
```

Sample Input:

10 15
7
OpenAI
16
Suchita
Welcome

Sample Output:

Sum of 10 and 15 is: 25
Area of circle with radius 7.0 is: 153.94
Reverse of 'OpenAI' is: IAnepO
Square of 16 is 256, Square root is 4.0
Hello, User!
Hello, Suchita!
Welcome, Suchita!

Note:

- The code for input taking and output display is given to you; write the logic inside the functions to achieve the task.
- The square root value in the output is rounded to 2 decimal places.

Source Code:

function_operations_dynamic.py

```
import math

def add_numbers(a, b):
    return a+b
# complete the remaining part of the code here...
def area_of_circle(radius):
    return math.pi*radius*radius
# complete the remaining part of the code here...

def reverse_string(s):
    return s[::-1]
# complete the remaining part of the code here...

def square_and_root(n):
    return n**2,math.sqrt(n)
# complete the remaining part of the code here...

def greet(name="User", greeting="Hello"):
    return f"{greeting}, {name}!"
# complete the remaining part of the code here...

a, b = map(int, input().split())
radius = float(input())
s = input()
num = int(input())
name_input = input()
greet_input = input()
```

```
# --- Output Display ---
print(f"Sum of {a} and {b} is:", add_numbers(a, b))
print(f"Area of circle with radius {radius} is:", round(area_of_circle(radius), 2))
print(f"Reverse of '{s}' is:", reverse_string(s))
sq, rt = square_and_root(num)
print(f"Square of {num} is {sq}, Square root is {round(rt, 2)}")
print(greet()) # Default
print(greet(name_input)) # With name
print(greet(name=name_input, greeting=greet_input))
```

Execution Results - All test cases have succeeded!

Test Case - 1
User Output
10 15
7
OpenAI
16
Suchita
Welcome
Sum of 10 and 15 is: 25
Area of circle with radius 7.0 is: 153.94
Reverse of 'OpenAI' is: IAnep0
Square of 16 is 256, Square root is 4.0
Hello, User!
Hello, Suchita!
Welcome, Suchita!

Test Case - 2
User Output
5 3
1
Python
9
Alex
Hi
Sum of 5 and 3 is: 8
Area of circle with radius 1.0 is: 3.14
Reverse of 'Python' is: nohtyP
Square of 9 is 81, Square root is 3.0
Hello, User!
Hello, Alex!
Hi, Alex!

Test Case - 3
User Output
-10 15
0
abc

4
John
Hey
Sum of -10 and 15 is: 5
Area of circle with radius 0.0 is: 0.0
Reverse of 'abc' is: cba
Square of 4 is 16, Square root is 2.0
Hello, User!
Hello, John!
Hey, John!